Discrete Drainage Dynamics: A Bounded, Conservative Local Rule for Newtonian-Like Potentials

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April 2026

Abstract

We define a deterministic, parallel, local update rule on a discrete graph. Each node carries a non-negative reserve $R_i \geq 0$, with rest value R_0 at equilibrium. Each oriented link carries a non-negative directional flux. The rule alternates a node update and a link update, with the outgoing fluxes from a node rate-limited so that the reserve cannot become negative. The flux on a link is written in a general form involving a *mobility function* $\mu(R)$ which encodes how the source node's reserve modulates its emission rate. The present paper adopts the constant-mobility baseline $\mu \equiv 1$, the simplest choice consistent with the substrate's locality and conservation properties. In this baseline, far from the non-negativity floor, the rule's steady state coincides with the discrete Poisson equation, and we measure on the lattice that the static profile around a point source follows a $1/r$ law with the continuum coefficient at the percent level. The contribution of this paper is the rule and the mobility framing, not the $1/r$ law. Companion papers study other mobility prescriptions and their non-linear consequences.

1 What this paper claims

We make three claims.

1. We define a deterministic, parallel, two-step local rule on a discrete graph. The reserve is bounded below by zero ($R_i \geq 0$, the physical floor: one cannot drain what is not there); R_0 is the rest value, but the reserve is free to exceed it in excited regimes treated in companion papers. The flux on a link involves a mobility function $\mu(R)$ that we leave as a structural choice of the framework; the present paper studies the constant-mobility baseline $\mu \equiv 1$.

2. Run as a stationary linearised simulation with a point source on a cubic lattice, the rule (with $\mu \equiv 1$) produces a radial profile compatible with $1/r$ in the bulk, with a coefficient that matches the discrete continuum prediction at the percent level across $L \in \{64, 128, 160\}$.

3. The internal flux-exchange rule preserves total reserve exactly. Non-negativity of the reserve is enforced by rate-limiting the outgoing fluxes and external sinks of a node in proportion, so that emission stops when the reserve reaches zero; no mass is created or destroyed by the internal rule. This is the structural feature that distinguishes the rule from a plain discrete heat equation. (The full rule contains a dormant gauge sector with its own invariants, not used in this paper.)

We do not claim the following. We do not derive Newton's gravitational constant in physical units; we calibrate it. We do not derive general relativity. We do not derive any property of the Standard Model. We do not show that the rule is unique. We do not derive the mobility function $\mu(R)$; the constant-mobility baseline of this paper is a choice within a family of possibilities. We do not assert that the $1/r$ result is itself novel: it is not. What is novel is the rule, the mobility framing, and the structural features (reserve non-negativity, rest level R_0, two-step alternation, rate-limited fluxes) that make the rule non-trivial in the regimes explored in the companion papers.

2 The picture, in plain words

Imagine a fine three-dimensional grid. At each grid point sits a reservoir holding a non-negative amount of stuff. There is a rest level R_0 that the reservoirs all sit at when the system is undisturbed; reservoirs can be excited above R_0 or drained below it, but they cannot go below zero. Between neighbours run pipes that carry a non-negative flow of stuff from the fuller reservoir to the emptier one, at a rate proportional to the level difference, optionally modulated by how much stuff the source actually holds. A reservoir cannot deliver more stuff than it actually has: a node at zero stops emitting. There is no separate "maximum flow rate" parameter; the cap follows from what the reservoir contains.

If we start the system at the rest level R_0 everywhere, drain one point, and let the system settle, the bulk reservoir levels relax into a stationary pattern. With the simplest choice of mobility ($\mu \equiv 1$, the case of this paper), far from the source the pattern reduces to the textbook discrete Poisson solution and the deficit $R_0 - R_i$ falls as $1/r$ with distance. Other choices of mobility yield qualitatively different non-linear regimes, studied in companion papers.

The interesting cases are where the floor matters. Near a strong drainage point, the reservoir level can be driven to zero, and emission from that node stops. In this regime the rule is no longer a discrete heat equation: it is a non-linear update with a finite-resource bound. The companion papers explore those regimes (Schwarzschild-form behaviour in Paper II, cosmogenetic regime in Paper III, gauge sector in Papers V–VI). The present paper is restricted to the linearised regime with $\mu \equiv 1$, where the rule reproduces $1/r$.

The rest of this paper makes the rule precise.

3 The substrate

3.1 Graph and variables

The substrate is a graph $G = (V, E)$ in which every node has at least six neighbours and the connectivity is statistically isotropic at large scale. We do not specify the exact topology; in our simulations we use a regular cubic lattice for clarity, and we have verified that BCC gives the same large-scale spectral dimension.

For the purposes of this paper, each node $i \in V$ carries a non-negative reserve $R_i \geq 0$, with rest value R_0 at equilibrium. Each oriented link $(i, j) \in E$ carries a pair of non-negative directional reserve fluxes $F_{i\to j} \geq 0$ and $F_{j\to i} \geq 0$. These are the only variables that enter the linearised gravitational simulation reported here. The non-negativity $R_i \geq 0$ is the physical floor (one cannot drain what is not there) and is the only structural source of non-linearity in the rule. In the static gravitational regime treated below with $\mu \equiv 1$, R stays at or below R_0 everywhere because the only source acts as a sink; in excited regimes (companion papers) R can exceed R_0.

Variables of the wider rule

The full rule, as used in the companion papers, also assigns to each node two angular quadratures (T_i, I_i) subject to a local coherence bound $T_i^2 + I_i^2 \leq R_i$, and to each oriented link a phase

$\psi_{ij} \in [-\pi, \pi)$ with $\psi_{ji} = -\psi_{ij}$. These additional variables drive the gauge sector (Papers V–VI), couple to the reserve through the coherence bound, and do not enter the static linearised gravitational regime treated below. We mention them here for completeness; we do not invoke them in the simulation or in the results of this paper.

3.2 Parameters

The rule has the parameters listed in Table 1. They are inputs.

Parameter	Role	Used in this paper
κ	drainage coefficient	yes
α	flux propagation coefficient	yes
R_0	reserve rest value (equilibrium)	yes
$\mu(R)$	mobility function (structural choice)	yes ($\mu \equiv 1$)
J	link relaxation coupling	no (gauge sector, deferred)

Table 1: Parameters of the local rule. The reserve floor at 0 is not a parameter: it is the physical statement that one cannot drain what is not there. R_0 is the rest level at equilibrium, not an absolute ceiling: the reserve can exceed R_0 in excited regimes treated in the companion papers. The mobility $\mu(R)$ is a structural choice of the framework: this paper uses $\mu \equiv 1$, companion papers (notably Paper II) use other choices.

4 The two-step tick

Motivation

The class of rules considered here is motivated by an old line of inquiry on whether physics admits a microscopic deterministic, local, iterative description — a "computational substrate" in the sense of Zuse, Fredkin, 't Hooft, Wolfram, and Lloyd [1–5]. We do not commit to the strong form of this hypothesis. We adopt it as a working stance: it tells us what shape a candidate substrate rule should have — deterministic, local, parallel, conservative, with bounded transport — and the present paper studies one such rule. The numerical claim of the paper does not depend on the working stance.

The rule

The rule decomposes into two alternating half-steps, which we call *Node Update* and *Link Update*. The choice of two-step alternation is a standard one for local lattice theories: it preserves locality (each update reads only from nearest neighbours) and avoids self-reference within a tick. We do not derive the alternation from a deeper principle; we adopt it as a construction.

4.1 Link Update

During the first half-tick, the node reserves are frozen. For each oriented link (i, j), in parallel, we compute a desired flux. We write the desired flux in the general form

$$F^{\mathrm{des}}_{i\to j} \;=\; \alpha\,\mu(R_i)\,(R_i - R_j)_+, \tag{1}$$

where $\mu(R)$ is a non-negative *mobility function* encoding how the available reserve at the source node limits its ability to push reserve into a neighbour. We normalise $\mu(R_0) = 1$ at the rest level, so that α retains its meaning as the rated propagation coefficient when the source is at rest. We do not derive $\mu(R)$ from a deeper principle; it is a structural choice of the rule, fixed by physical considerations specific to the regime under study. Different choices of μ yield different non-linear steady-state equations.

Constant-mobility baseline of this paper. The present paper adopts the constant-mobility baseline,

$$\mu(R) \equiv 1, \tag{2}$$

for which Eq. (1) reduces to the textbook discrete-diffusion form $F^{\text{des}}_{i\to j} = \alpha(R_i - R_j)_+$, and the linearised steady state is the discrete Poisson equation, giving the $1/r$ profile measured in Sec. 6. This is the simplest choice consistent with the substrate's locality and conservation properties.

Other mobility choices in companion papers. The reserve-proportional mobility $\mu(R) = R/R_0$, motivated by local bandwidth limitation at the source node, is studied in Paper II, where it is shown to produce $\nabla^2(R^2) = 0$ in the continuum limit and a profile of Schwarzschild form. Other mobility prescriptions (double-throttle, power-law, or more elaborate forms) may be explored in subsequent papers as needed. Within Paper I, none of these is invoked.

The two directions of a link are independent. Each node then computes a rate-limiting factor that ensures the reserve stays non-negative on its outgoing side. Define the available local budget and the desired total demand,

$$A_i \;=\; R_i + \tau\, S_i^{\text{in}}, \quad S_i^{\text{in}} = \sum_j F^{\text{des}}_{j\to i}, \quad D_i \;=\; \tau\, O_i^{\text{des}} + \tau\,\kappa\, E_i, \quad O_i^{\text{des}} = \sum_j F^{\text{des}}_{i\to j}, \tag{3}$$

where the demand D_i counts *both* the desired total outflow and the local sink, since both draw from the same reserve. The rate-limiter is

$$\beta_i \;=\; \begin{cases} 1, & D_i = 0, \\ \min(1,\; A_i/D_i)\,, & D_i > 0, \end{cases} \tag{4}$$

and is applied identically to the outflows and to the sink:

$$F_{i\to j} \;=\; \beta_i\, F^{\text{des}}_{i\to j}, \qquad E_i^{\text{real}} \;=\; \beta_i\, E_i. \tag{5}$$

When demand exceeds the available budget, $\beta_i < 1$ scales outflows and sink down in the same proportion. When the reserve is comfortably above zero, $\beta_i = 1$ and the rule is linear.

4.2 Node Update

During the second half-tick, the link fluxes are frozen. For each node i, in parallel,

$$R_i \;\leftarrow\; R_i + \tau\, S_i^{\text{in}} \;-\; \beta_i\, \tau\, O_i^{\text{des}} \;-\; \beta_i\, \tau\, \kappa\, E_i. \tag{6}$$

By construction of β_i in Eq. (4), this update keeps R_i non-negative at every tick, without any clipping by hand. The internal exchange of reserve between nodes is exactly conservative; the external sources E_i enter as explicit bookkeeping terms.

4.3 Why this is not the discrete heat equation

In the regime where every node has a comfortable available budget, $\beta_i = 1$ everywhere, and the rule reduces (with $\mu \equiv 1$) to a discrete heat equation. Outside that regime, it does not: a node whose available local budget $A_i = R_i + \tau S_i^{\text{in}}$ falls short of its desired demand D_i has $\beta_i < 1$, and its outflows and sink are scaled down in proportion. A node at zero reserve with no incoming compensation stops emitting altogether. The desired outflow is not realised; the reserve does not become negative; no mass is created or lost. This non-linearity is not present in the standard discrete heat equation. It is the structural feature that the companion papers exploit (Paper II for Schwarzschild-form behaviour with non-trivial mobility, Paper III for cosmogenetic regimes).

In terms of the deficit $\delta_i = R_0 - R_i$, the desired flux in the constant-mobility baseline reads $F^{\mathrm{des}}_{i\to j} = \alpha\,(\delta_j - \delta_i)_+$: the reserve flows from the fuller site toward the emptier one.

In the static gravitational regime treated below, where $R \le R_0$ everywhere, the bound on the flux satisfies $0 \le F_{i\to j} \le \alpha R_0$. This is a downstream consequence of the initial conditions and source type, not a separate cap imposed on the rule.

5 Conservation

Total reserve, exactly conserved by internal exchange. The actual realised fluxes $F_{i\to j} = \beta_i F^{\mathrm{des}}_{i\to j}$ enter the Node Update of j as $+\beta_i F^{\mathrm{des}}_{i\to j}$ (incoming) and the Node Update of i as $-\beta_i F^{\mathrm{des}}_{i\to j}$ (outgoing). The two contributions cancel when summed over the whole graph, so $\sum_i R_i$ is exactly conserved by the inter-node flux dynamics. The rate-limiter β_i does not violate conservation: it scales outflows and the local sink in the same proportion, so any reduction of an outgoing flux at i is matched by a reduction of the absorbed energy $E^{\mathrm{real}}_i = \beta_i E_i$, never by a creation or destruction of reserve.

External sinks as bookkeeping. The total reserve in the network changes only by $\sum_i E^{\mathrm{real}}_i$, which counts what is taken out by external sources. The closed system consisting of the network plus the external sink reservoir is conservative; the network alone is open, and the sink terms are explicit accounting of that openness, not a violation of conservation.

Dormant gauge sector. The full rule contains a link-phase ψ_{ij} whose closed-loop circulation modulo 2π is preserved by the topology. This invariant is not active in the gravitational regime studied here.

6 Simulation: from rule to $1/r$

6.1 Derivation of the Poisson limit (constant mobility)

We work here in the constant-mobility baseline $\mu \equiv 1$, where the desired flux is $F^{\mathrm{des}}_{i\to j} = \alpha(R_i - R_j)_+$. In the linear regime $\beta_i = 1$ everywhere, and the desired fluxes coincide with the realised fluxes. The two directional fluxes on a link satisfy

$$F_{i\to j} - F_{j\to i} \;=\; \alpha(R_i - R_j)_+ - \alpha(R_j - R_i)_+ \;=\; \alpha(R_i - R_j),$$

since exactly one of the two terms is non-zero. The net inflow into node i is therefore

$$S_i = \sum_{j\sim i}(F_{j\to i} - F_{i\to j}) = \alpha\sum_{j\sim i}(R_j - R_i) = \alpha\,\Delta R_i,$$

where $\Delta f(i) = \sum_{j\sim i} f(j) - n_i f(i)$ is the discrete graph Laplacian (with $n_i = 6$ on cubic). Writing the Node Update (6) in stationary regime and dividing by τ,

$$\alpha\,\Delta R_i \;=\; \kappa\,E_i. \tag{7}$$

Switching to the deficit $\delta_i = R_0 - R_i$, since R_0 is constant we have $\Delta\delta_i = -\Delta R_i$, and

$$-\alpha\,\Delta\,\delta_i \;=\; \kappa\,E_i. \tag{8}$$

For a single point source $E_i = E_0\,\delta_{\mathrm{Kr}}(i, i_0)$, this is the discrete Poisson equation

$$-\Delta\,\delta_i \;=\; M\,\delta_{\mathrm{Kr}}(i, i_0), \qquad M \;\equiv\; \frac{\kappa\,E_0}{\alpha}, \tag{9}$$

with source strength M set by the imposed sink magnitude E_0 and the microscopic ratio κ/α. The continuum solution is $\delta(r) = M/(4\pi r)$ since $\Delta(1/r) = -4\pi\,\delta^{(3)}(r)$ in 3D.

6.2 Numerical check

We use a cubic lattice with absorbing (Dirichlet) boundary conditions, iterate to convergence by Jacobi relaxation, bin the field radially, fit $\delta(r) = A/r + B$ over the bulk shell $3 \le r \le 0.3L$, and extract A at three lattice sizes $L \in \{64, 128, 160\}$.

L	A_{fit}	relative bias	statistical precision	N_{iter}
64	0.07878	-1.01%	7.0×10^{-5}	5 000
128	0.07866	-1.16%	5.7×10^{-6}	10 000
160	0.07886	-0.90%	3.7×10^{-6}	30 000
$A_{\text{continuum}} = M/(4\pi)$		0.07958		

Table 2: Newton-fit coefficients on cubic lattices of three sizes, $M = 1$, Dirichlet boundary, with $\mu \equiv 1$. The fit reproduces the $1/r$ form. The bias is roughly L-independent at the percent level, consistent with the discrete-lattice Madelung correction at short range, not a boundary effect in $1/L$.

6.3 Robustness: spectral dimension

We compute the spectral dimension $d_s(t)$ of the graph Laplacian on cubic and BCC at $L \in \{32, 64, 128\}$ via the heat-kernel return probability $P(t) = \langle e^{-\lambda t} \rangle$, with $d_s(t) = -2\,\mathrm{d}\log P/\mathrm{d}\log t$. The plateau value of d_s converges to 3 as L grows.

L	cubic ($d_s - 3$ minimum)	BCC ($d_s - 3$ minimum)
32	0.129	0.107
64	0.054	0.045
128	0.023	0.019

Table 3: Spectral-dimension residual converges to 0 in the IR limit on both substrates. The gravitational sector is Newtonian in the IR limit on this class of substrate.

6.4 Code and data availability

The Python scripts used to reproduce Tables 2 and 3 are available at https://github.com/stanislasdewavrin/DDD-Paper-1-Foundations. The scripts are pure-numpy and do not depend on any external simulation framework.

7 Calibration vs. prediction

Calibration. The mapping between the dimensionless deficit field δ_i and the physical Newtonian potential Φ, and the mapping between the local energy budget E_i and physical mass, are external calibrations: in this paper, δ and E remain dimensionless lattice quantities. The lattice unit of length and the parameter κ are fixed by demanding that the coefficient A_{fit}, when matched to Newton's law, reproduces the measured G for a chosen test mass. This is one calibration condition for two parameters, and it fixes a one-parameter family of physical realisations.

Prediction. Within that family, the rule (with $\mu \equiv 1$) predicts that the static potential of an isolated point mass is $1/r$ at distances on which the linearised regime is valid, and that finite-resource departures from $1/r$ appear when the bounds become active.

We separate these two cleanly throughout the companion papers. "DDD agrees with X" means prediction; "DDD is calibrated against Y" means calibration.

8 Model checks and extension-level consequences

The table below collects three different kinds of statement that are sometimes lumped together and that we keep separate. It mixes (i) numerical checks internal to the simulation of this paper, (ii) consequences expected of the rule that are worked out in companion papers, and (iii) physical falsifiers that would apply once the rule is calibrated to physical units. We label which is which.

Consequence	Regime	Type	Failure mode / test
$1/r$ static potential	isolated mass, linear regime, $\mu \equiv 1$	numerical check (this paper)	deviation from $1/r$ above bin-statistical noise in the bulk fit window.
Maximum signal speed c_{eff} tied to αR_0 and lattice spacing	all dynamic regimes	structural consequence	faster-than-c_{eff} propagation in simulation; mismatched c from EM and gravitational wavefronts after calibration in the companion paper.
Non-linear mobility corrections to $1/r$	$\mu(R) \neq 1$, static spherical regime	extension-level prediction (Paper II)	absence of the predicted R^2-harmonic profile under the bandwidth-throttled mobility.
Departure from $1/r$ when $R_i \to 0$	near-source, very high deficit	extension-level prediction (saturation)	absence of any cutoff in simulations, or in observed compact-object phenomenology.
Time-reversal asymmetry of U_{tick}	two-step alternation	structural consequence	exact time-reversal symmetry of any prediction that depends on the alternation order.

Table 4: Model checks and consequences expected of the rule, by type.

9 Relation to existing work

The single-point claim of this paper, that a local rule produces a $1/r$ static potential, is by itself trivial: any diffusive process in three dimensions has the same property, and the discrete graph Laplacian's Green function is a textbook result. The contribution of the paper is the *rule* and its mobility framing, not the result. We position it briefly against neighbouring literatures.

Lattice gauge theory [6]. The two-step alternation of node and link updates mirrors the standard matter-gauge structure of Wilson lattice gauge theories, and the link-phase ψ_{ij} in the wider rule plays the role of a U(1) gauge variable. The distinction here is that the link sector carries a directional reserve flux $F_{i\to j}$, automatically bounded by the non-negativity of the reserve. The presence of a finite-resource floor on a propagating quantity, plus a mobility function as a structural choice, is not standard in lattice gauge theory and is what the companion papers exploit.

Cellular automata for physics [3,4]. Local deterministic rules on discrete substrates have a long history. Most CA programmes target unification of multiple sectors of physics from a single substrate. The present paper is more modest: we define one rule (with one mobility choice) and one regime, and we do not claim more.

Causal sets and discrete-spacetime gravity [7,8]. These programmes treat the substrate itself as dynamical (the graph or simplicial complex evolves). Here the substrate is fixed; the dynamics live on its node and link variables.

Discrete diffusion and graph Laplacian [9]. The static linear regime of our rule is the discrete Poisson equation, and the $1/r$ result is its direct consequence. We do not claim novelty here. The contribution is the embedding of this linear regime inside a non-linear, bounded, two-step rule, together with a mobility framing that admits non-trivial steady-state equations for non-constant μ. The numerical result in Sec. 6 should accordingly be read as a validation of the implementation under the constant-mobility baseline, not as a new theorem about graph Laplacians.

Entropic and thermodynamic gravity [10, 11]. These derive Newton's law from thermodynamic identities on horizons. The present rule is a microscopic mechanical model, not a thermodynamic derivation. The two are not in tension; they target different levels of description.

Lattice fluid models [12]. Lattice Boltzmann methods are conservative, local, parallel updates of density and momentum fields with caps and floors built in for stability. The structural similarity to the present rule is real, and we use lattice-Boltzmann-style implementation techniques in the simulation. The physical interpretation is different: lattice Boltzmann targets fluid mechanics, the present rule targets a substrate from which gravitational and gauge-like behaviour are intended to emerge.

10 What the model is not, and what it does not do

The rule is Newtonian in its present static linearised form with $\mu \equiv 1$. We do not derive general relativity in this paper, and the companion paper on the weak-field limit (Paper II) is a weak-field limit obtained under a different mobility choice, not a derivation of the full Einstein equations.

We do not derive Newton's gravitational constant in physical units; we calibrate it. A research programme to derive it is sketched in Paper II.

We do not derive the fine-structure constant, particle masses, or any property of the Standard Model.

We do not specify the exact graph topology beyond "minimum 6 neighbours, statistically isotropic". We have tested cubic and BCC, and they agree at the percent level.

We do not give a microscopic origin for the reserve rest value R_0, which sets the structural scale of the rule. The floor at zero is fixed by the physical statement that one cannot drain what is not there.

We do not derive the mobility function $\mu(R)$. The present paper studies the constant-mobility baseline $\mu \equiv 1$, for which the linearised steady state is the discrete Poisson equation. Different mobility choices lead to different non-linear steady-state equations; the constant-mobility choice is the minimal Newtonian baseline. Companion papers study specific reserve-dependent mobility prescriptions, in particular the source-side bandwidth-throttled choice $\mu(R) = R/R_0$ in Paper II.

We have not shown that the rule is unique. There may exist other local, conservative, bounded rules that produce a $1/r$ profile in their linearised regime.

11 Open questions

Why six neighbours? Six neighbours is the minimal coordination of the cubic stencil used in our simulation; it is not a fundamental requirement of the rule. More general graphs require separate checks of spectral dimension and isotropy. We have no microscopic argument for why the substrate would prefer any specific coordination.

Why this order, Node then Link? The two orderings are not equivalent: U_{tick} is not invariant under exchange. This is a microscopic breaking of time-reversal. We have no microscopic argument for the choice.

What sets R_0? R_0 is the rest reserve level at equilibrium, the only structural scale of the rule. The floor at zero is fixed physically. We do not derive R_0.

What sets the mobility $\mu(R)$? The constant-mobility baseline $\mu \equiv 1$ used in this paper is the simplest choice. Other choices (in particular the source-side bandwidth-throttled $\mu(R) = R/R_0$

studied in Paper II) yield qualitatively different non-linear regimes. A microscopic argument that selects a specific mobility from the rule of Paper I would close an important gap of the framework.

Is the rule unique? See Sec. 10.

What sits below the substrate? The graph is the lowest level of description here. Whether it is reducible to anything more fundamental is beyond the present paper.

12 Outlook

The companion papers explore further consequences of the same local rule with different mobility prescriptions.

1. Paper II — weak-field gravitational behaviour with the source-side bandwidth-throttled mobility $\mu(R) = R/R_0$, yielding the Schwarzschild time-dilation factor. *Linearised plus saturation; partial.*

2. Paper III — discrete-time response and inertial effects. *Speculative.*

3. Paper IV — predictions and tests of the gravitational sector. *Mixed.*

4. Paper V — the gauge sector (T, I, ψ) and emergent electromagnetism. *Speculative; coefficients uncalibrated.*

5. Further exploratory notes (Papers VI and onward) address coupling, cosmology, quantum phenomenology, particle structure, and condensed-matter analogies. *Increasingly speculative.*

The reader should expect the support for our claims to weaken as the series progresses. The present paper, Paper I, claims only the rule itself (with constant-mobility baseline) and the linearised $1/r$ verification. The rest is what one might do with such a rule.

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